chatserver.

```
import java.io.*;
import java.util.*;
public class chatserver {
    static set < printWriter > clients =
    Collections.synchronized set (new Hashset<>();
    public static void main (string[] args)
throws exception {
while (true {
    Datagram Packet P = new Datagram Packet (b, b.leng
    S.receive (P);
    S.send (new Datagram packet (r, r.Length,
            P.get Address(), p.getPort()))
}
S.close();
```

Chat client.java.

```
import java.io.*;
public class chatclient {
public static void main (string [] args) throws
Exception {
    Socket S = new Socket ("107.0.0.1", 806);
new thread (() -> {
    try {
        string m;
        while ((m = in.readline() != null)
    }
    S.close();
}
```

Scanned with OKEN Scanner

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PING PROGRAM

AIM:

To implement a user defined ping program to check the reachability of a host & obtain its response

Algorithm:

1. Start the program
2. Get the hostname or IP address from the user
3. Execute a ping request to the specified host
4. Wait for response from the host
5. If a reply is received, display the host is reachable & response details
6. Stop the program

PROGRAM:

```
import java.io.*;
public class MyPing {
    public static void main(String[] a)
throws Exception {
    BufferedReader b = new BufferedReader
(new InputStreamReader(System.in));
```

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CODE WITH RAW SOCKETS IMPLEMENT PACKET SNIFFING

AIM :

To write a python programming using a raw socket to capture network packets and display the source MAC address, protocol number & packet Length.

Algorithm :

1. Start the Program
2. Import the socket and struct Modules
3. Create a Raw socket using AF-Packet for packet-level access.
4. Continuously receive Packet from the network
5. Extract the ethernet reader from each Packet
6. IPV4 packets, extract the source and destination address
7. Diplay the simulation
8. Close the socket

PROGRAM :

```
import socket
raw_socket = socket.socket(socket.AF_PACKET)
```